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Massive recycling of waste mobile phones: Pyrolysis, physical treatment, and pyrometallurgical processing of insoluble residue

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KEYWORDS: *Waste mobile phone (WMP), Precious metals, Pyrolysis, Physical treatment, Smelting, Slag*

ABSTRACT: Waste mobile phones (WMP) consisting of a heterogeneous mixture of metal, plastic, glass and ceramic materials have either disposal or recycling problems. Pyrolysis can prevent the release of dioxin because organic materials are decomposed during the heating process. This paper reports a three-step massive treatment of WMP by pyrolysis, physical treatment, and pyrometallurgical processing. The release of dioxin was significantly limited during pyrolysis. Miscellaneous solid parts after pyrolysis (= 488 kg/ton-WMP) were physically separated into char (= 181 kg/t-WMP), iron scrap (= 75 kg/t-WMP), printed circuit boards (PCBs = 121 kg/t-WMP), and insoluble residue (= 111 kg/t-WMP). Here, the smelting of insoluble residue was carefully investigated to simulate the recovery of precious metals (gold and silver) and critical metals (nickel and tin) in a molten state. Recovery rate of valuable elements was influenced by terminal velocity of metallic particles in the liquid slag in association with slag viscosity and silicate structures. The comprehensive metal recovery system can produce substantial amounts of copper (82.7 kg), gold (0.1 kg), silver (0.3 kg), nickel (1.5 kg), and tin (3.3 kg) from the processing 1000 kg of WMP.

INTRODUCTION

Waste of electrical and electronic equipment (WEEE) has diversified due to the rapid development of the information technology (IT) industry. Specifically, a significant amount of mobile phones are thrown away because technical trends lead to the faster replacement of goods.[1,2] Reuse or repair of mobile phones have long been decline due to rising product complexity and shortened life cycles.[3] Unlike general waste, waste mobile phones (WMP) include various heavy metals and hazardous substances. Lead, cadmium, mercury, PVC, and halogenated flame retardants are examples of those that pollute the environment.[4] Mobile phones also contain significant amounts of resource materials derived from precious metals, carbonaceous materials, and inorganic parts. Valuable metals can be recovered from the lead frame (copper), solder (lead and tin), case (iron and aluminum) and IC chip (gold, silver, and palladium), which particularly increases the recycling rate of mobile phones.[5] Among the various parts, printed circuit boards (PCB) are considered to be essential to the urban mine industry due to integrated platinum group metals (PGMs).[6] The global recycling of waste PCBs is mainly concentrated in Asia. The current recycling status of PCBs in China and South Korea were reviewed in recent publications.[4,7]

There have been several studies on the recovery of metals from electronic waste [8-11] and waste PCBs.[5,6,12-17] Recycling techniques can be divided into either chemical or physical treatment, with the non-metallic fraction (NMF) treated separately.[14] Metallurgical recovery of metals from electronic waste is classified as hydrometallurgical processing or pyrometallurgical processing depending on the operating temperature.[9] Metal recovery by the hydrometallurgical process is considered to be more exact, more predictable and more easily controlled than the pyrometallurgical process.[9]

The most active research on the recovery of metals from electronic scrap concerns the recovery of precious metals by hydrometallurgical techniques.[6,18-20] The main steps consist of a series of acid or caustic leaches of the waste material. The solutions are then subjected to separation and purification procedures,

such as precipitation of impurities, solvent extraction, adsorption, and ion-exchange, to isolate and concentrate the metals of interest.[20] Process is followed by an electro-refining, chemical reduction, or crystallization for metal recovery.[22]

Although the hydrometallurgical processes have been the traditional methods of recycling WEEEs, a large amount of waste acid liquid and sludge are produced, which increases the waste management costs.[16] PCBs are manufactured in various types and sizes, ranging from single-layered to multi-layered, and single-sided to double-sided, and the components placed on each PCB can also be variant both in function and in materials.[14] In the manufacturing of a PCB, conducting layers of thin copper foil and insulating dielectric composite fibers are deposited. It is difficult to physically separate these components even in the finely ground condition. The efficiency of leaching is extremely low in this instance, therefore, the flow of recycling various metals from waste PCBs becomes long and complicated and the recovery cost becomes very high.[16]

Pyrometallurgical treatment can be used for the mass processing of waste PCBs regardless of their manufactured type.[23] Recent investigations using pyrometallurgical fundamentals have been reported by researchers.[13,24-29] However, there exists a concern on the incineration of electronic equipment due to the release of toxic substances such as heavy metals, brominated flame retardants (BFRs), dioxin, and dibenzofurans.[15] Atmospheric pollution is even worse in the case of PCBs as the copper incorporated acts as a catalyst for dioxin formation when flame retardants are incinerated.[9,30] In this regard, thermal recycling techniques by pyrolysis have been widely researched as a method of recycling synthetic polymers including polymers that are mixed with glass fibers.[15] Vacuum pyrolysis combined with mechanical processing [15] and a method of using centrifugal separation for solders [16] are examples of prominent pyrometallurgical processes.[13] Pyrolysis of WEEE can be operated at relatively low temperatures (400-600 °C) to minimize the release of dioxin or furan. The volume of WEEE is not only reduced to 60% during pyrolysis but an easier separation of PCBs from electronic equipment is attained by further physical separation.[31]

Smelting processes to extract precious metals from PCBs by adding automotive catalysts were developed by Kim *et al*.[25] Gold, palladium, and platinum were extracted from raw materials by accumulation at a Cu-Sn-Fe alloy. The recovery of PGMs was higher than 90% according to the report.[25] Kwon *et al.*[32] investigated the recovery of valuable metals in terms of the slag viscosity and size of crushed PCBs. The viscosities of the 45%CaO-40%SiO_2-15%Al_2O_3 oxide flux system were measured as about 0.5 Pa·s at 1500 °C and 3.1 Pa·s at 1300 °C. However, slag viscosities obtained from PCB scrap fluxed by an oxide system were 0.3 Pa·s at 1500 °C and 7.0 Pa·s at 1300 °C. Increased slag viscosity at 1500 °C was due to the various impurities within the waste PCBs.[26] Insoluble residues in PCBs mainly composed of oxides are responsible for the complicated slag design. Slag basicity and viscosity are significantly influenced by residues that change the recovery rate of precious metals.[33] Previous pyrometallurgical studies [24-26] did not cover the release of toxic substances, and the recent pyrolysis studies have not considered the insoluble residue which significantly impedes the leaching process.[10,16,17,31]

In the present study, three-step method; pyrolysis, physical treatment and pyrometallurgical processing was recommended for the recycling of WMP. The benefits of the proposed methodology are obvious as listed below.

(i) The release of dioxin is minimized by the pyrolysis.

(ii) The metal (Cu, Sn, and Ni) recovery can be maximized through the physical separation.

(iii) Smelting process can effectively extract precious metals from insoluble residues.

The smelting treatment of insoluble residue was mainly covered to simulate the recovery procedure of precious metals in the molten state. Slag composition was controlled by incorporating different compositions of synthetic fluxes. Recovery rate was significantly influenced by the terminal velocity of metal particles in the liquid slag in association with slag viscosity and silicate structures.

EXPERIMENTAL SECTION

Collected mobile phone scrap was tested for pyrolysis by the pilot scale rotary kiln. As an industrial cooperation, more than a hundred trials were conducted in an effort to increase the amount of scrap in the furnace and decrease the amount of hazardous gas emissions. Pyrolysis of WMP of 1000 kg was carried out at 500 °C and release of dioxin was measured by gas mass spectrometer (JMS-700d, JEOL, Japan). Mechanical separation was conducted subsequently. Crushed samples were classified by vibrating screening and iron scrap was collected by magnetic separator. Ground samples were separated by gravity separator to classify char, PCBs, and insoluble residue. Insoluble residues and PCB were quantitatively analyzed by inductively coupled plasma atomic emission spectroscopy (ICP-AES, OPTIMA 8300, PerkinElmer, Waltham, USA).

The smelting experiments for insoluble residue were carried out by using a high-frequency induction furnace with a graphite heating element. A schematic diagram of the experimental apparatus is shown in Figure 1.

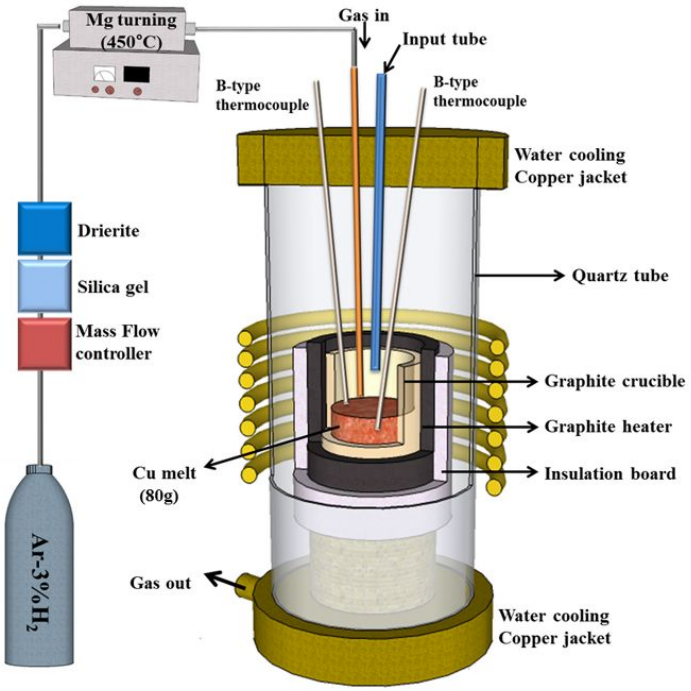


Figure 1. Schematic of the high-temperature induction furnace used in the smelting process

The quartz reaction chamber was initially evacuated using a mechanical rotary pump and then filled with an Ar-3%H_2 gas mixture. The temperature was controlled within ±2°C using a B-type (Pt-30wt%Rh/Pt-6wt%Rh) thermocouple and a proportional integral differential (PID) controller. Type B thermocouple was used for the temperature calibration according to the reference table of platinum at 1200°C (International Temperature Scale, ITS-90). A highly purified Ar (99.999%) gas was supplied during the experiments by a mass flow controller. Minor oxygen in the gas was removed by passing the gas through magnesium turnings heated to 500°C. Residual impurities in the Ar gas were removed by passing the gas through Drierite (W.A. Hammond Drierite Co. Ltd., Xenis, OH, USA) and silica gel. The graphite crucible was selected to exclude the influence of refractory erosion on final slag composition. Chemical reaction by the carbon was considered as negligible since CaO-Al_2O_3-SiO_2 oxide system was difficult to be reduced at 1500 °C. Insoluble residue of 100 g was used for each experiment to estimate the recovery of the precious metals. Preliminarily fused CaO-SiO_2-Al_2O_3 ternary slags (30 g) were used with different ratios of CaO to SiO_2 as listed in Table 1.

Table 1. Oxide Composition of Insoluble Residue and Designed Fluxes

	Contents (wt%)		
	SiO_2	Al_2O_3	CaO
Insoluble residue	42.5	40.0	17.5
Flux A	61.0	5.0	34.0
Flux B	52.0	5.0	43.0
Flux C	43.0	5.0	52.0

Copper (99.99%) granules of 80 g were initially melted at 1500 °C for 30 minutes to produce a metal pool which plays a role as a valuable metal absorber. The mixed raw materials (insoluble residue 100 g, copper 80 g, and flux 30 g) were smelted for 2 hours until the completion of reactions. The specimen was quenched by flushing with pure Ar gas (99.999%). The slags and metals were carefully separated from the crucible. The slag was finely ground to powder, and the metal was cut into small pieces for chemical analysis.

The metal contents are obtained by inductively coupled plasma - atomic emission spectroscopy (ICP-AES, OPTIMA 8300, PerkinElmer, Waltham, USA). Before analysis, an acid leaching of the sample with aqua regia was carried out for 12 hours for the complete dissolution of elements. Then, part of the digestion solution was diluted with ultrapure distilled water.[30] For each analyzed element, the mean concentration was averaged from three measurements. The composition of the slag was analyzed with X-ray fluorescence spectroscopy (XRF, S4 Explorer; Bruker AXS Inc., Madison, USA). Metal droplets entrapped in the slag phase were investigated by using the backscattered electron (BSE) SEM (Model JSM-6380LA, JEOL, Tokyo, Japan) equipped with energy dispersive X-ray spectrometry (EDX, 20 keV, SUTW Sapphire detector, Mahwah, NJ) with ZAF quantification. More 50 points were analyzed and normalized to represent the compositions of copper droplets.

■ RESULTS AND DISCUSSION

Pyrolysis, Physical Separation and Smelting of Insoluble Residues. The flow process of WMP treatment; Pyrolysis, physical separation, and pyrometallurgical treatment is shown in Figure 2. The maximum amount of WMP for a mass pyrolysis was increased to 1000 kg by means of pilot scale rotary kiln. The consumption of natural gas (157 m^3/ton-WMP) and electricity (156.9 kWh/ton-WMP) were optimized through hundreds of trials. In particular, the amount of dioxin generated during the pyrolysis process was as low as 0.039 ng-TEQ/Sm^3, which was measured by mass spectrometer (JMS-700d, JEOL, Japan).[35] The regulative limitation for dioxin in exhaust gas in South Korea is reported as 0.1 ng-TEQ/Sm^3.[36] A schematic gas treatment system for possible oil separation is also shown in Figure 2.

Physical separation of the specimen was carried out after pyrolysis. The products were classified as char (18.0 wt%), iron scrap (7.5 wt%), PCB (12.1 wt%), and insoluble residue (11.1 wt%) as a result of magnetic and size separation as shown in Figure 2. Char was separated from insoluble reside by gravity separation through shaking table. Char and iron scrap are readily used by the steel industry to make foundry

products. Precious and base metals are mainly recovered from PCBs, and insoluble residue. The chemical analysis of PCBs and insoluble residues by ICP-AES is shown in Table 2.

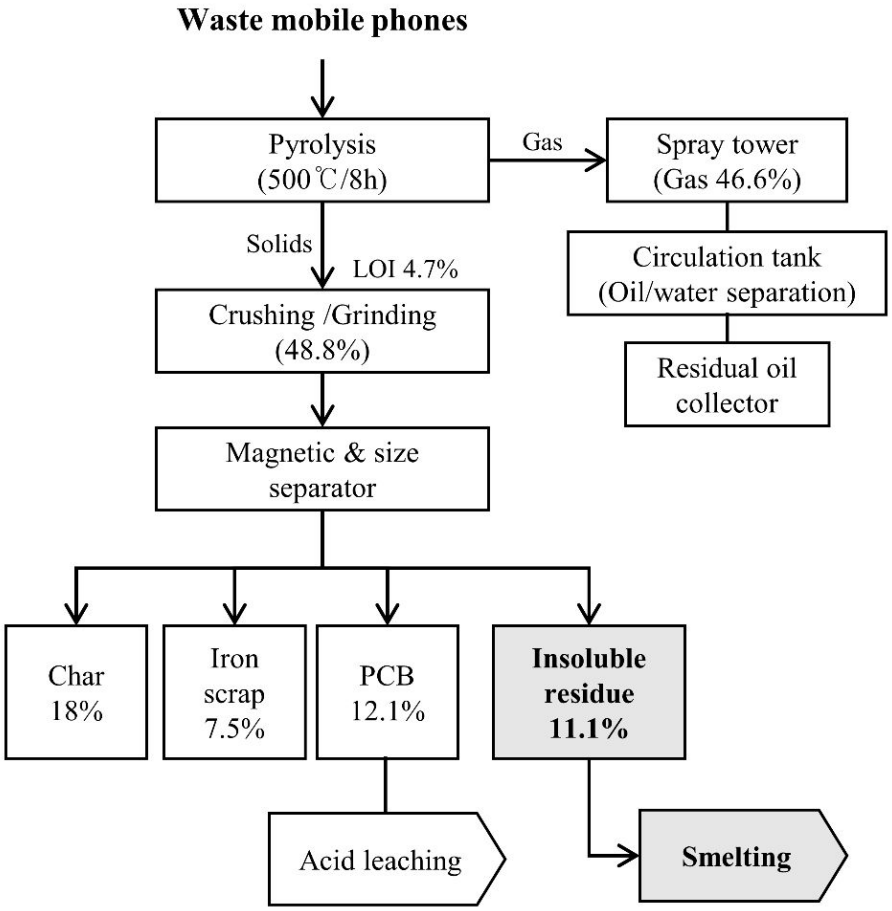


Figure 2. Flow process based on unit weight of spent mobile phone treatment; Pyrolysis, physical separation and smelting.

Table 2. Composition of Metal Elements in Insoluble Residues and PCBs Analyzed by ICP-AES

	Content (wt%)										
	Cu	Fe	Al	Sn	Ni	Pb	Zn	Ag	Au	Pd	Total
Insoluble residue	9.3	-	-	0.27	0.33	-	-	0.11	0.04	0.01	10.06
PCBs	59.9	-	0.87	2.46	0.91	-	-	0.15	0.05	-	64.34

There was only 9.3 wt% copper in the insoluble residue and more than 90 wt% was composed of oxides. However, because gold, silver, nickel and tin contents in the insoluble residue were as high as PCB parts, these elements must be recovered to increase the profit of the overall recycling process.

Leaching efficiency of precious metals from insoluble material is significantly reduced due to the ceramics that cover metallic parts. Oxides such as SiO_2, Al_2O_3, and CaO physically interfere with the acid leaching of precious metals from insoluble residues.[17] Disassembly and physical separation of the metal component layers in PCBs is even more difficult when crushed insoluble residues are encountered as shown in Figure 3.[6] Thus, copper, nickel, silver, and gold need to be recovered by smelting process to separate metals from oxides.

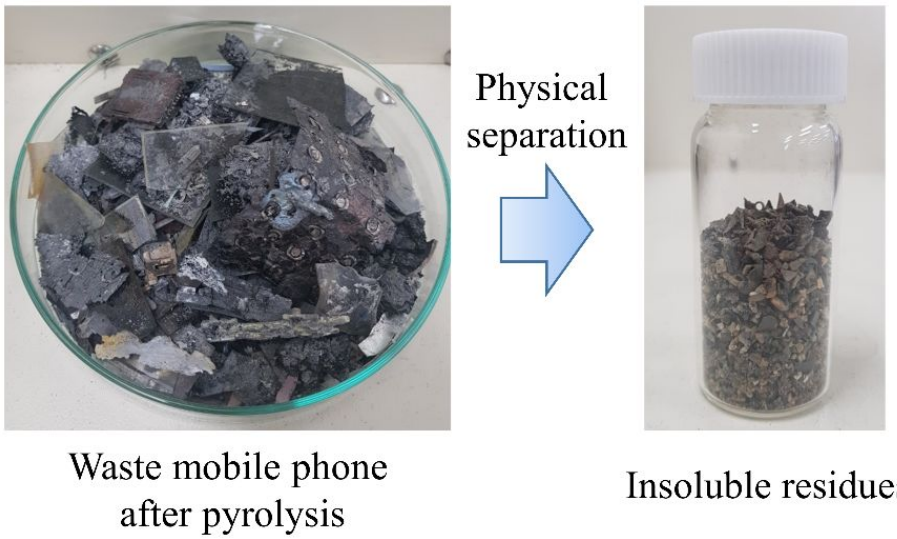


Figure 3. WMP after pyrolysis and insoluble residues after physical separation.

The existing oxides in PCBs prevent smelting process due to the high melting temperature of SiO_2 (≅ 1710 °C) and Al_2O_3 (≅ 2070 °C) compared to the low melting temperature of copper (≅ 1080 °C). A flux

system that effectively decreases the melting temperature of molten slag and easily formulates low viscous slag needs to be considered. The initial oxide composition of insoluble residue and three flux candidates with different CaO/SiO_2 ratios, called basicity index, are illustrated in Figure 4.

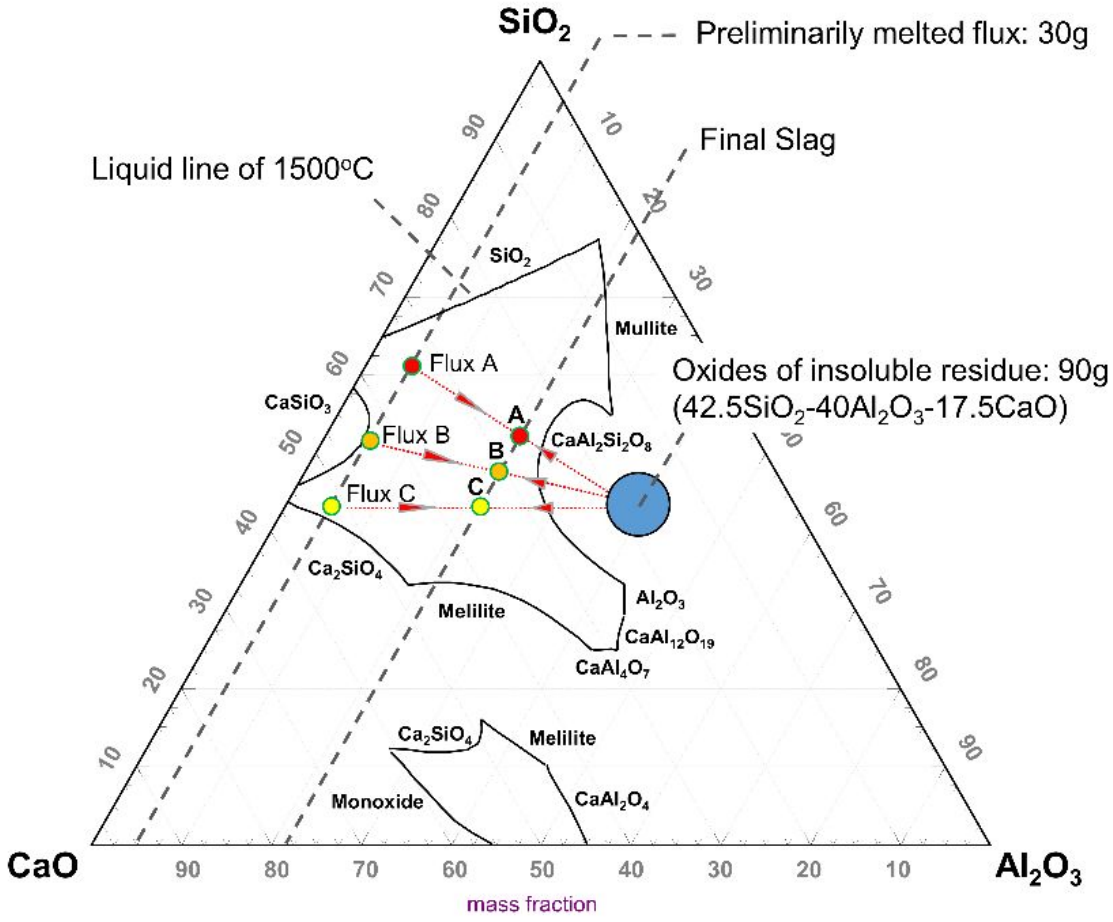


Figure 4. Oxide composition of insoluble residue, different fluxes (A, B, and C) and prospected slag compositions illustrated in the CaO-Al_2O_3-SiO_2 ternary phase diagram at 1500 °C.

Obtained slag compositions depending on varied fluxes are indicated on each linear line because the raw materials are merged at a fixed quantity in the liquid phases. As indicated in Figure 4, melting temperature of oxide of insoluble residue (42.5%SiO_2-40%Al_2O_3-17.5%CaO) is higher than 1500 °C.

Fluxes only combined with CaO, SiO_2, and 5wt% of Al_2O_3 is mandatory that does not change the major composition of slag system. The compositions of obtained slags (A, B, and C) are all included in liquid line of 1500 °C. Minimal amounts of fluxes (30 g) were confirmed which has the minimal compositional change and sufficiently low temperature (1500 °C) for melting. The separation between copper metal and slag were observed from all specimens.

Metal Recovery of Smelting Process. After the smelting experiment, a quenched sample was obtained. The metal parts mainly composed of copper were cleanly separated and weighed (90.4 g). Since 80 g of Cu was used as a liquid metal absorber, increased weight accounted for Cu, Au, Ag, Ni, and Sn collected from insoluble residues. Precious metals (Au, Ag) and critical metals (Ni, Sn) dissolved in copper were measured by ICP-AES and are indicated in Table 3.

Table 3. Concentration of PGM and Base Metals Incorporated in Metal and Slag Phases Analyzed by ICP-AES

	Metal Phase (ppm)				Slag Phase (ppm)				
	Au	Ag	Ni	Sn	Au	Ag	Ni	Sn	Cu (wt%)
Slag A	490	1040	8570	3900	330±4	70±2	570±23	440±16	0.41
Slag B	390	980	8510	4040	50±2	60±5	150±4	150±6	0.19
Slag C	540	1100	10670	5070	70±3	70±1	190±10	180±9	0.16

The glassy slag was collected and ground for quantitative chemical analysis. Basicity index did not change after smelting and the Al_2O_3 incorporated in the slag was fixed at approximately 21% as expected. The copper content of the slag is crucial as copper metal droplets have a significant influence on metal recovery rate.[33,34] The remaining coppers in smelting slag were measured to be 0.41% in Slag A, 0.19% in

Slag B, and 0.16% in Slag C as listed in Table 3. Twice the amount of copper was confirmed in Slag A compared to the other two samples. The copper content has a strong relationship with slag viscosity associated with physical separation. High silica content in the slag A significantly increased the viscosity and confined copper droplets to the slag. The concentrations of Au, Ni, and Sn in Slag A are thus much higher than in Slags B and C due to the behavior of copper in slags. A strong relationship between the behavior of copper and other metals was confirmed.

In the copper smelting process, for instance, precious metals and impure metals, such as Pb, Bi, Sn and As, is closely connected to copper loss in slag because those metals have high solubility in molten copper.[39] Precious metals are well known recovered by-products in copper smelters.[40] Gold in a copper concentrate is absorbed in copper matte during flash smelting, and then the gold is collected as a residue after electro-refining of the copper anode. Highly pure gold can be obtained by further electro-refining.[40] In the present experiment, gold that is dissolved in the copper melt has a similar chemical behavior to copper.[29,41] Nickel also has high mutual solubility with copper as they form a homogeneous solid solution in any mixing ratios.[41] Tin is an excellent alloying element for copper and has been used since the bronze ages.[41] Consequently, targeted metals for recovery, such as gold, nickel, and tin, behave similar to a liquid solution of copper, therefore they are enormously affected by the physicochemical property of liquid slag.

The influence of slag composition on metal recovery of smelting process was investigated by measuring dissolution of the metals in the slag systems. The recovery rate of metals is described by equation (1).

$$\text{Recovery rate (\%)} = \frac{[\text{wt\% } i]_{\text{metal}}}{[\text{wt\% } i]_{\text{metal}} + (\text{wt\% } i)_{\text{slag}}} \times 100 \quad (1)$$

where *i* is an element in the metal and slag phase such as Ag, Au, Ni, and Sn. The recovery rate of valuable elements for smelting process was calculated by using the analytical data of ICP-AES and is shown in Figure 5.

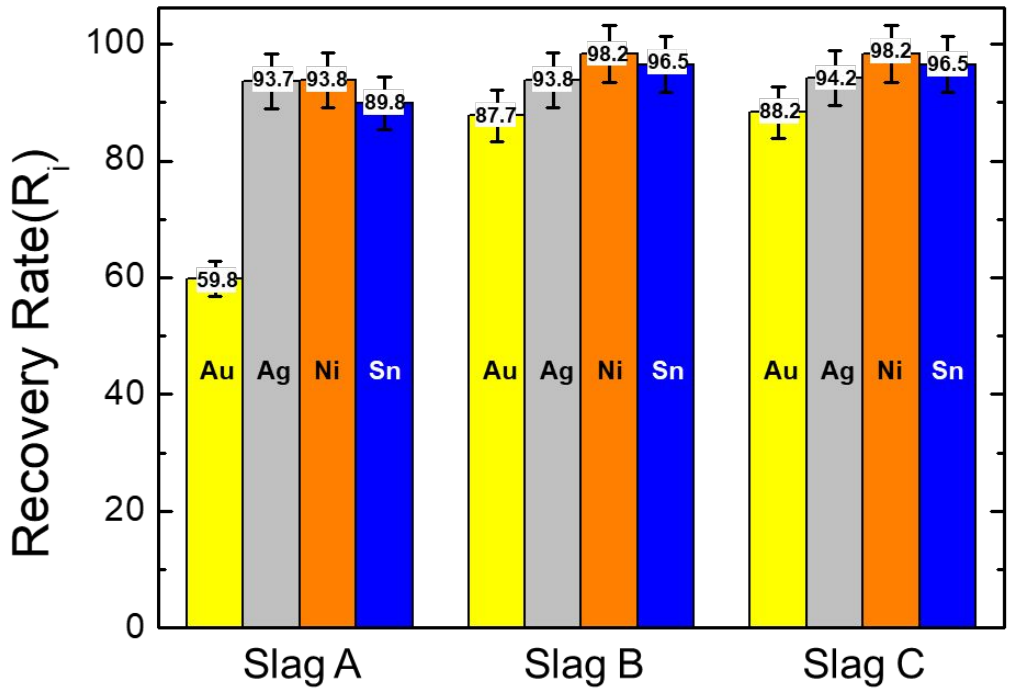
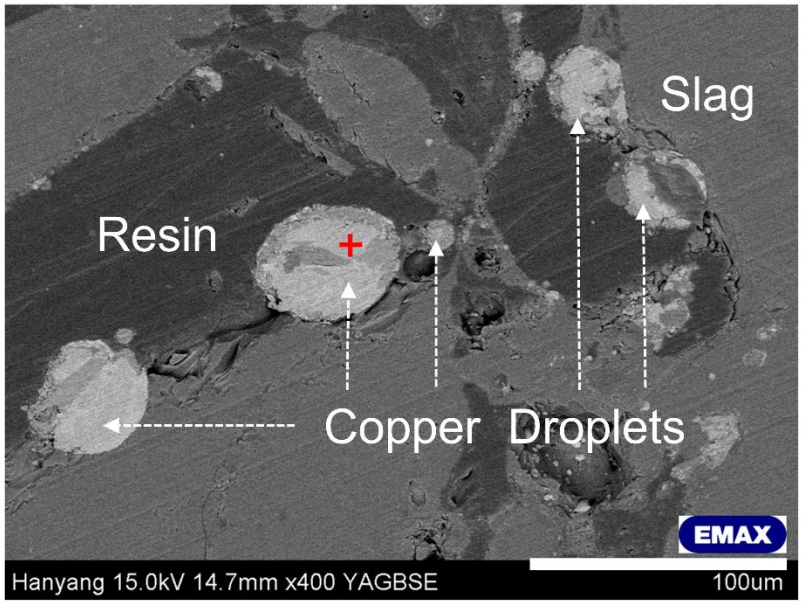


Figure 5. Recovery rate of Au, Ag, Ni, and Sn in Slag A, Slag B, and Slag C.

The recovery rates of Au, Ag, Ni, and Sn were similar in Slag B and Slag C while relatively lower recovery rates were observed in Slag A. The lowest recovery rate of gold was observed in Slag A (59.8%), whereas recovery rates were higher in Slag B (87.7%) and Slag C (88.2%). The recovery rate of Ag, Ni and Sn were slightly reduced in Slag A compared to Slags B and C. As alloying elements to copper, Ni and Sn behave with the same mechanism as liquid copper. The copper content in Slag A was much higher than the content in Slag B and Slag C as previously discussed in Table 3. The physical condition of copper droplets

in the slag will explain the phenomenon. The smelting process takes advantage of the gravitational separation of liquid metal and liquid slag. Physical separation accompanied by chemical reactions determines the efficiency of metal recovery. A high silica content in Slag A caused a significant decrease in the recovery rates of metals. The copper droplets confined to Slag A are shown in Figure 6, which indicates imperfect metal separation. Actually, the physical inclusion (entrainment) of copper droplets is the major reason for copper loss in commercial smelters.[37]



Element	Weight%
Si K	12.13
Ni K	12.29
Cu L	70.10
Ag L	0.13
Au M	5.34
Totals	100.00

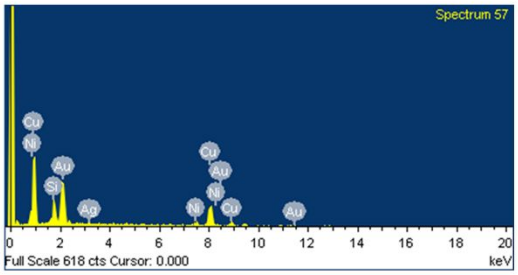


Figure 6. SEM image and EDX result of copper droplets confined to Slag A.

Influence of Physical Property of Slags on Metal Recovery. The recovery rates of Au, Ag, Ni, and Sn changed when fluxes with different basicity index were used. The physical impact of metal droplets on gravitational separation is as important as the chemical influence of flux that changes metal

solubility. The morphology of the slag sample was observed by SEM-EDX as shown in Figure 6. Metal droplets under 100 μm of diameter were confirmed as copper alloy. The minor elements, such as Au, Ag and Ni were observed by EDX (Si was excited from the slag matrix). Copper droplets including 0.5-5.0 wt% Au were detected, confirming a high gold dissolution in liquid copper. In Slag A, numerous metal droplets were confined in the slag phase due to the high slag viscosity as mentioned above.

It is considered that the technical removal of metal droplets in the slag phase is the most crucial factor that determines metal recovery rate. Thus, the terminal velocity of metal droplets in the slag was calculated in the present study. Metal droplets in the slag were assumed to be spheres. The relationship between the gravity, buoyancy and frictional forces on particles were considered. Assuming Stokes regime because of the low stirring in the slag, it can be described by Eq. (4).[42]

$$V_t = \frac{2r^2(\rho_m - \rho_s)g}{9\eta}, \quad \text{Stokes' law} \tag{4}$$

The terminal velocity of metal droplets in the slag is directly proportional to the square of the radius of metal droplets. In the present study, the slag viscosity was calculated by FactSage™7.0 (ESM Software, Hamilton, OH, USA), which is a commercial thermochemical computing software.[43] The viscosity of Slag A was 25.6 dPa·s (=Poise), while that of Slag B and Slag C were 16.7 and 8.5 dPa·s, respectively. The density of copper was used for the density of metal droplets because the major element in the metal droplets was copper. Slag density was referenced from the literature value.[44] However, the density gap between metal and slag ($\rho_m - \rho_S$) has little effect on terminal velocity as the density does not change significantly with the slag composition.

Terminal velocity of the metal droplets with various particle sizes (10, 50, and 100 μm) in the CaO-

SiO_2-Al_2O_3-5wt%MgO slag system were calculated using Eq. (4) as shown in Figure 7. Terminal velocity significantly decreases with increasing slag viscosity. For the comparison, a particle size of 100 μm in the copper smelting slag showed the fastest terminal velocity due to the lower viscosity. Copper particles had the lowest terminal velocity when the amount of silica was highest. The low recovery rate of metals in Slag A is thus explained by these physical property issues. Consequently, during the smelting process, the recovery of valuable metals from insoluble residue significantly affected by slag viscosity rather than by slag basicity. The physical properties of slag had the greater impact on the concentration of metals in the copper melt. Slag viscosity and associated terminal velocity of copper droplets need to be precisely controlled to maximize the recovery of valuable metals.

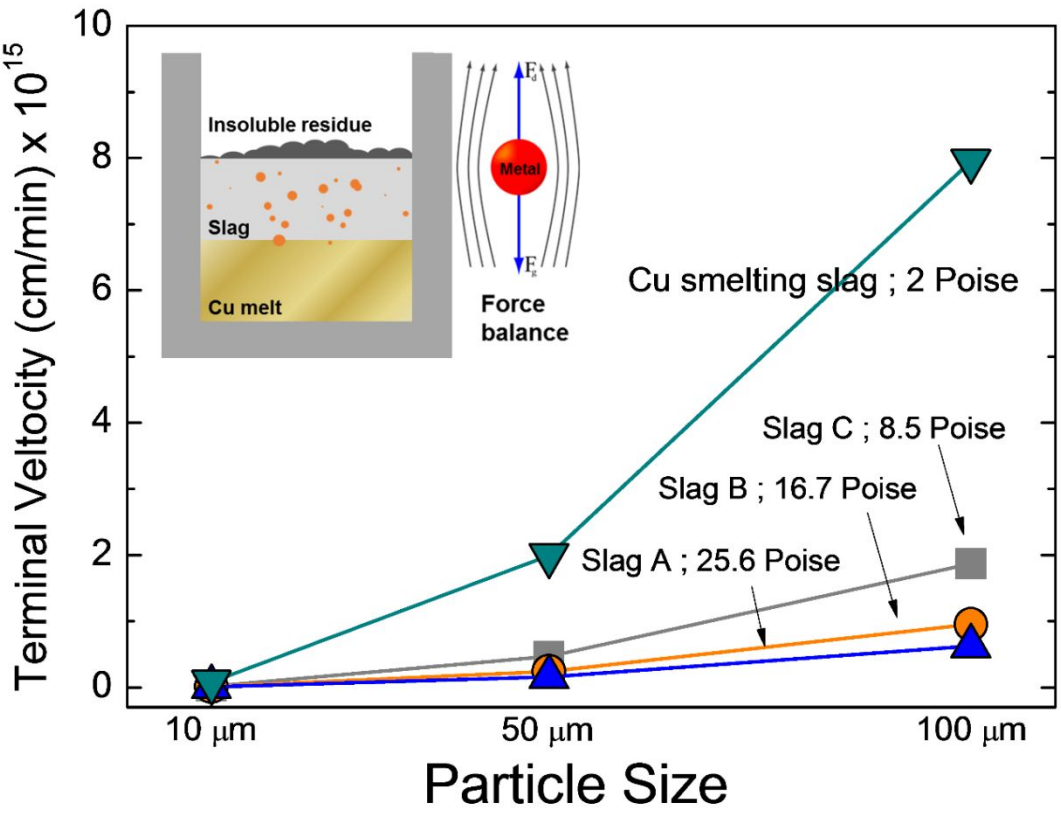


Figure 7. Calculated terminal velocity in slags depending on particle sizes and viscosity.

Material flow of pyrolysis – physical separation – smelting was described in graphical abstract by indicating mass balance of precious metals. PCBs (121 kg) and insoluble residues (111 kg) take the highest economic values compared to char (181 kg) and iron scrap (75 kg) due to contained metals. Cu, Au, Ag, Ni and Sn were assumed to be easily leached and recovered from PCBs. Recovered amounts of precious metals from insoluble residues and PCBs were calculated. Cu (82.7 kg), Au (98.6 g), Ag (297.1 g), Ni (1.5 kg), and Sn (3.3 kg) could be effectively recovered from WMP (1000 kg) though the methodology suggested in this study.

■ CONCLUSIONS

The comprehensive recycling system of pyrolysis – physical separation – smelting was confirmed to be efficient process for sustainable treatment of WMP. Massive amount of scraps (1000 kg) were processed with the pyrolysis with dioxin release lower than 0.039ng-TEQ/Sm^3. Physically classified materials; char, iron scrap, PCBs, and insoluble residue can be effectively utilized for material resources.

Smelting of insoluble residue was carried out to avoid inefficient acid leaching of oxide-complicated material. Synthetic flux was made to decrease the melting temperature and viscosity of molten slag. Recovery of precious metals of Au, Ag, Ni and Sn was directly related with copper droplet that forms in liquid slags. Since PGMs, Ni and Sn have high solubility in liquid copper, they are collected in copper alloy after smelting process. Behavior of copper droplets in the slag was described by terminal velocity as the index of recovery rate. Flux with high amount silica (34%CaO-61%SiO_2-5%Al_2O_3) showed the poor recovery rate of Au, Ni and Sn due to the high viscosity. Slag viscosity and associated terminal velocity of copper droplets is the primary factor to maximize the recovery of valuable metals.

■ ACKNOWLEDGMENTS

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■ NOMENCLATURE

V_t: Terminal velocity

r: Radius of metal droplets

ρ_m: Density of metal

ρ_s: Density of slag

g: Acceleration of gravity (= 9.8 m/s^2)

η: Viscosity of slag

PCB: Printed circuit board

PGM: Platinum group metal

WEEE: Waste of electrical and electronic equipment

WMP: Waste mobile phone

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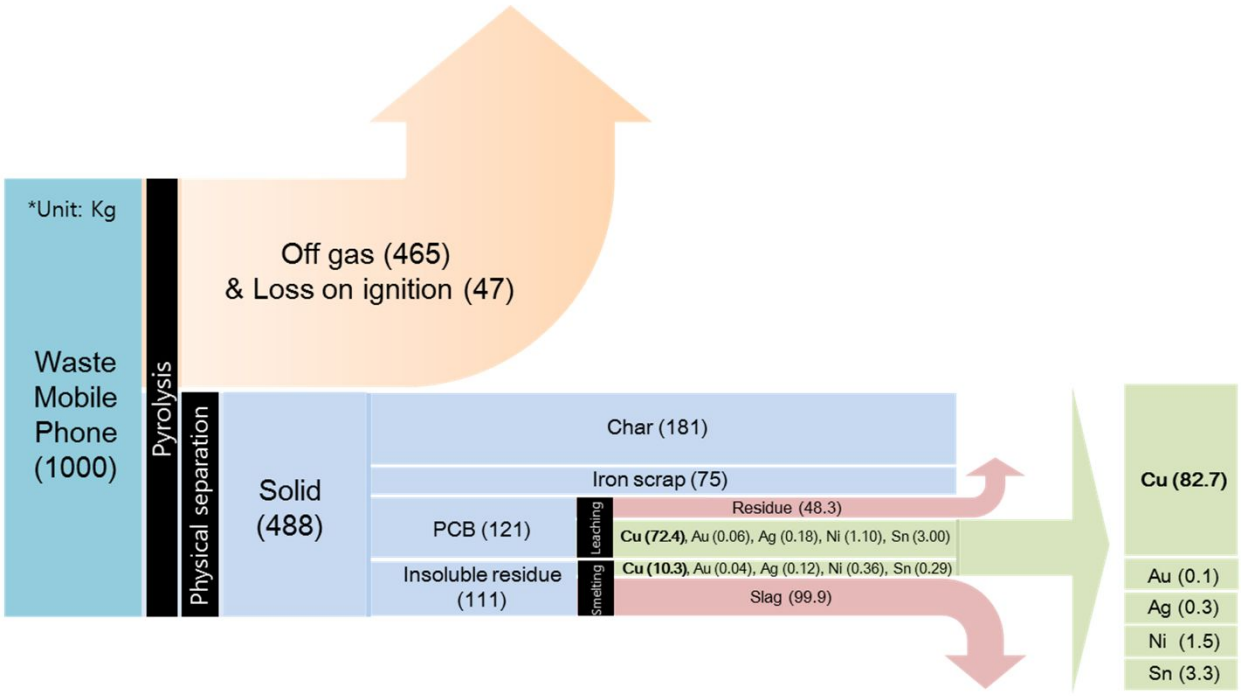
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TABLE OF CONTENTS (TOC) GRAPHIC



Synopsis

Comprehensive treatment of waste mobile phones: pyrolysis, physical separation and smelting processes have rendered the effective and sustainable metal recovery.

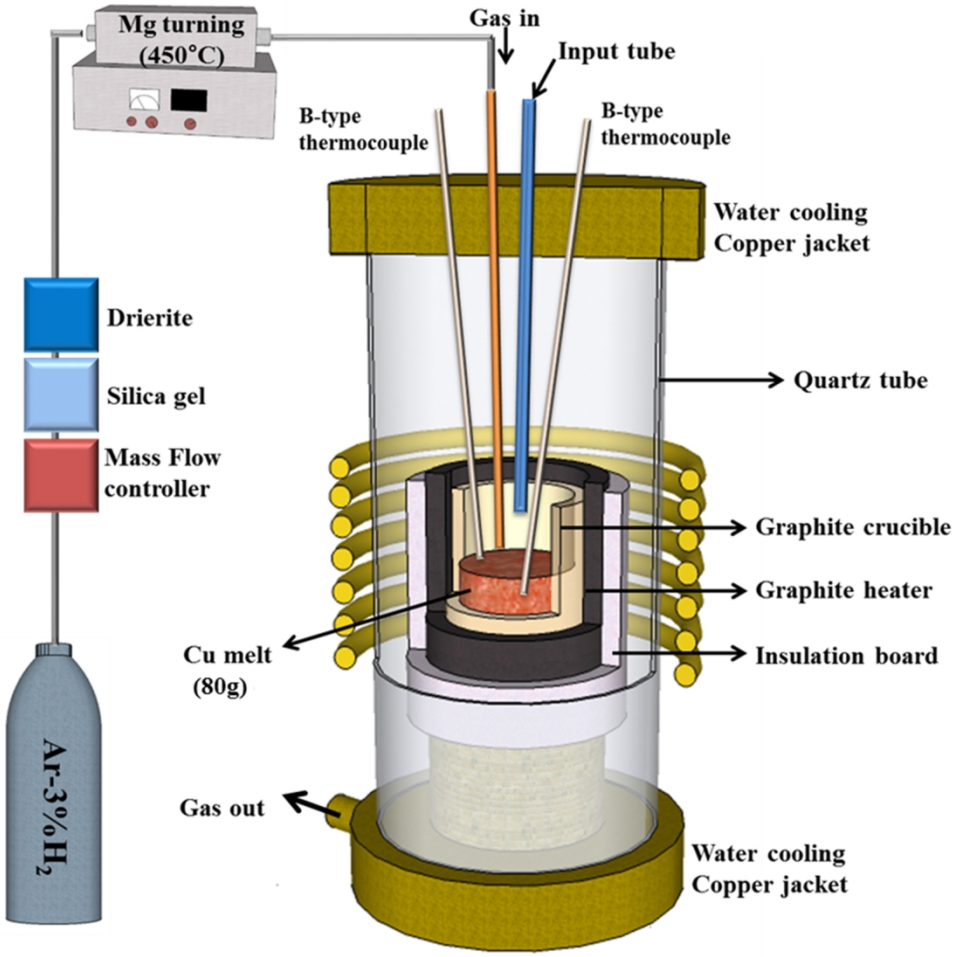


Figure 1. Flow process of spent mobile phone treatment; Pyrolysis and physical separation.

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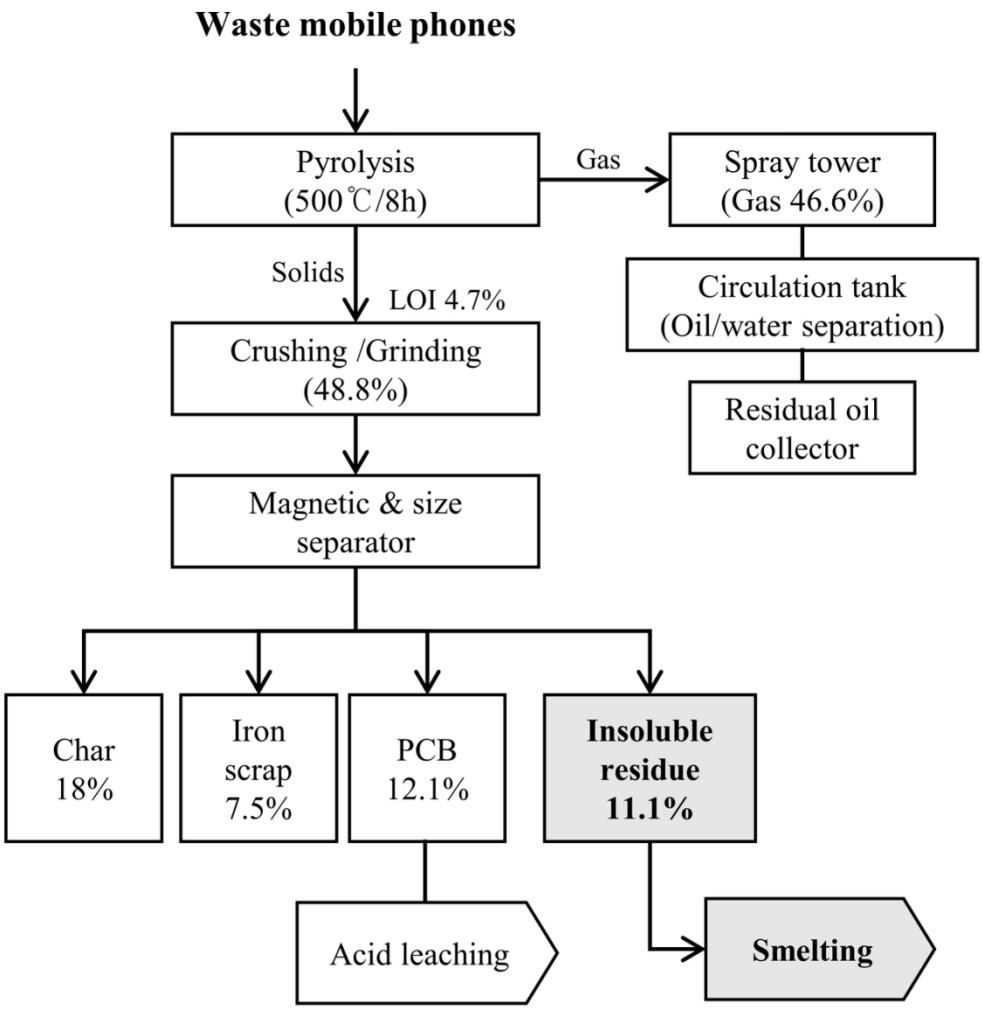


Figure 2. Flow process of spent mobile phone treatment; Pyrolysis and physical separation.

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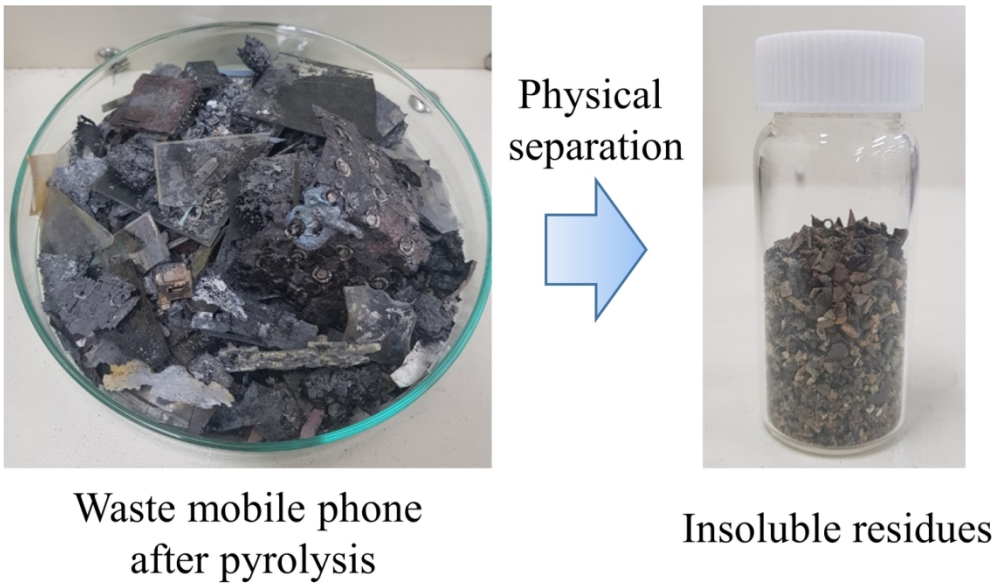


Figure 3. Waste mobile phone after pyrolysis and insoluble residues after physical separation.

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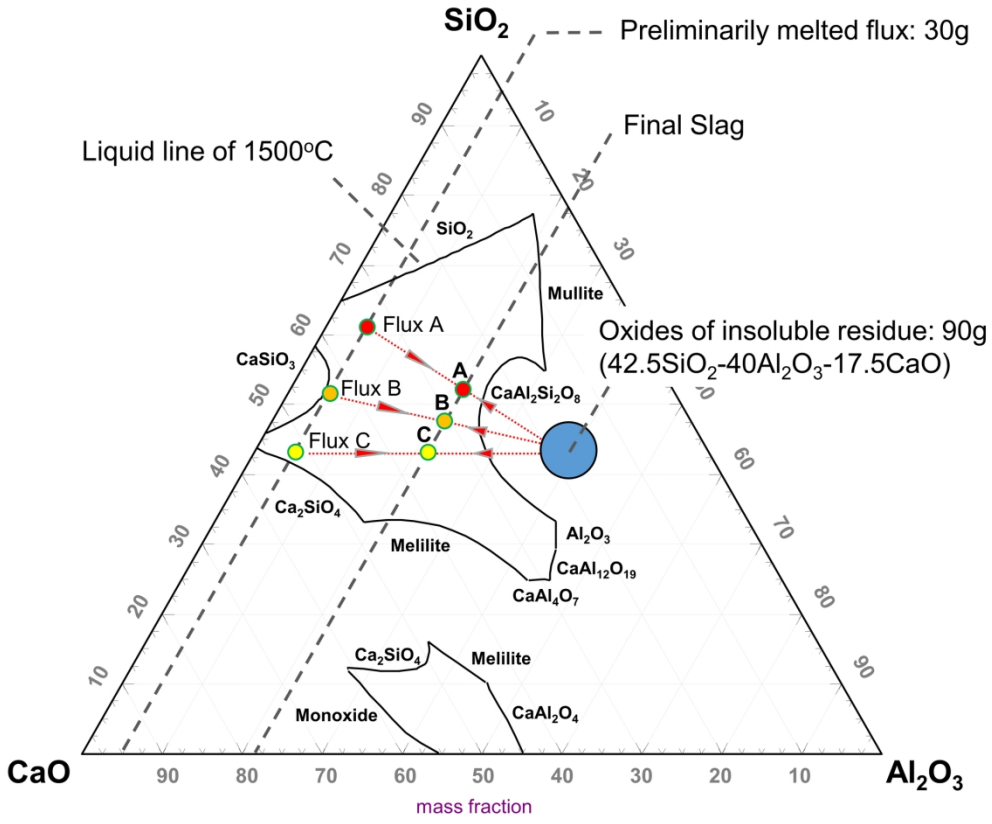


Figure 4. Oxide composition of insoluble residue, different fluxes (A, B, and C) and prospected slag compositions illustrated in the CaO-Al2O3-SiO2 ternary phase diagram at 1500°C.

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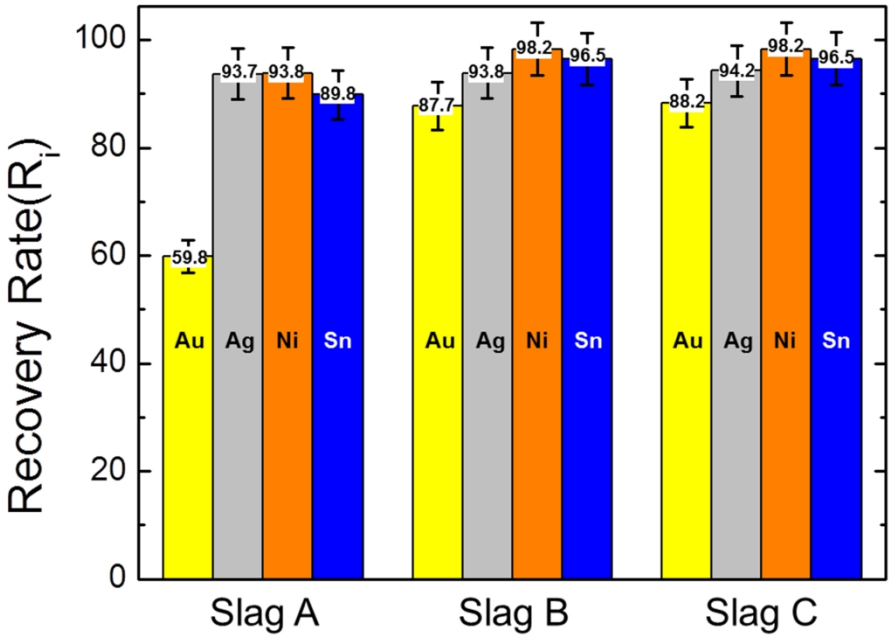
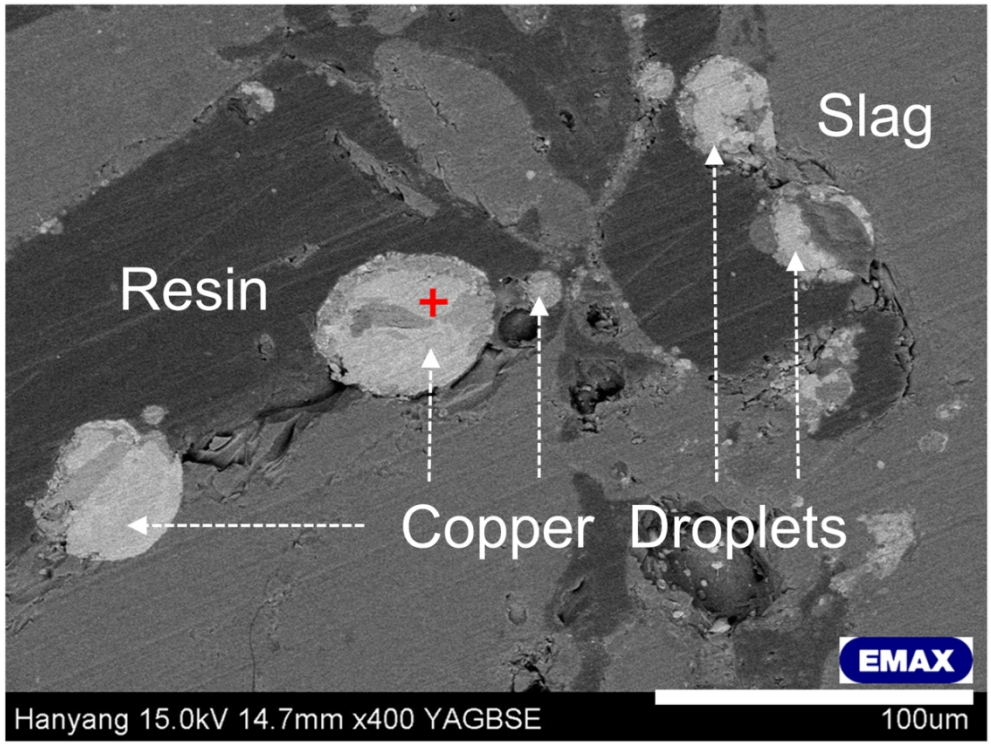


Figure 5. Recovery rate of Au, Ag, Ni, and Sn in Slag A, Slag B, and Slag C.

253x190mm (300 x 300 DPI)



Element	Weight%
Si K	12.13
Ni K	12.29
Cu L	70.10
Ag L	0.13
Au M	5.34
Totals	100.00

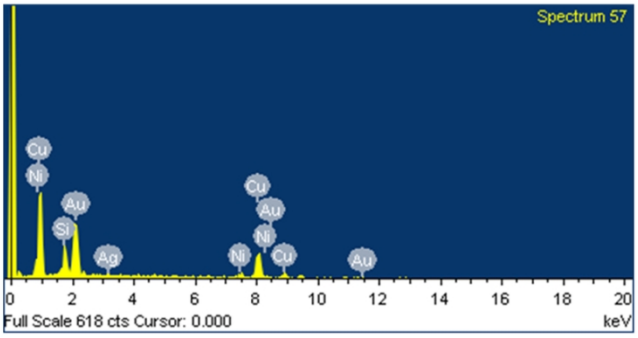


Figure 6. SEM image and EDX result of copper droplets confined to Slag A.

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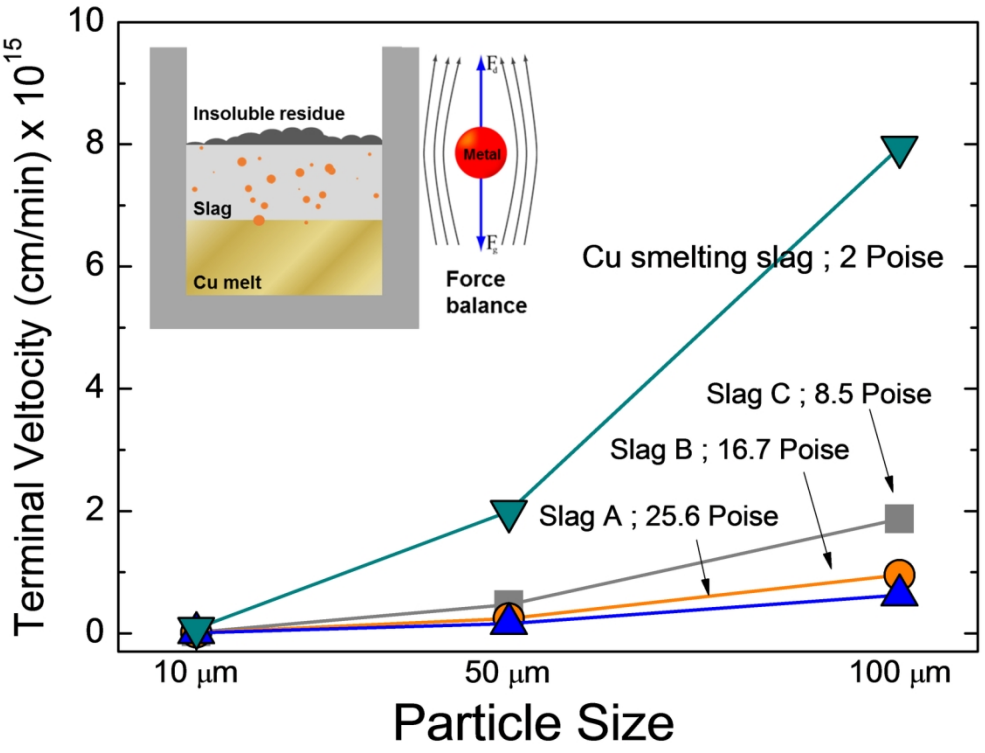


Figure 7. Calculated terminal velocity in slags depending on particle sizes and viscosity.

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